Automated Brain Tumor Segmentation
ECE 176 Final Project Report

Emmy Wei
Department of Electrical
and Computer Engineering
University of California, San Diego
A17949795

Terri Tai
Department of Electrical
and Computer Engineering
University of California, San Diego
A17876782

Abstract

Accurate localization of tumors in medical images is critical for diagnosis, treatment planning, and surgical guidance. However, automated segmentation remains challenging due to heterogeneous appearances, diffuse boundaries, and strong class imbalances between tumorous and healthy tissues. In this project, we investigate automated brain tumor segmentation from magnetic resonance imaging (MRI) using deep convolutional neural networks (CNNs). Using the BraTS dataset, we implement a 2D U-Net architecture that processes multimodal MRI inputs (T1, T1CE, T2, and FLAIR) to segment tumor regions at the pixel level. The model is trained using Dice Loss and evaluated using the Dice coefficient across tumor subregions. Our results demonstrate that U-Net can reliably localize tumor structures in MRI slices while maintaining stable training behavior even with limited data. This project highlights the practicality of deep learning-based segmentation methods for medical imaging applications.

1 Introduction

In neuro-oncology, accurately identifying the location and extent of brain tumors is critical for diagnosis, treatment, and monitoring. Traditionally, tumor segmentation is performed manually by radiologists, which is time-consuming and subjective. Our motivation is based on the fact that automated segmentation systems based on deep learning have the potential to significantly reduce the clinical workload of neurosurgeons and radiologists while improving consistency and reproducibility.

The primary goal of our project is to develop a deep learning model capable of automatically segmenting brain tumors from MRI scans. By leveraging CNNs, we aim to provide an automated tool that enables more precise measurements and localization of the tumor by performing pixel-wise classification, potentially reducing the time between diagnosis and treatment planning, while also helping with monitoring the efficacy of different treatments over time.

A major challenge in tumor segmentation is the large class imbalance problem, where the majority of pixels correspond to healthy tissue while tumorous tissues occupy only a small portion of the image. To address this issue, we utilize the Dice Loss function, which is suitable for our task because it is calculated based on the overlap between the predicted segmentation mask and the ground truth. This loss function is well-suited for our task specifically because it emphasizes the correct prediction of small targeted areas.

Additionally, another challenge is accurately capturing the complex shapes and diffuse boundaries of tumors. To handle this problem, we employ an encoder-decoder architecture like U-Net to do spatial feature extraction from MRI images, so we can capture the finer details of the tumor edges. We also use skip connections between the corresponding encoder and decoder layers to allow the network to have high-resolution spatial information.

Overall, there are three main clinically relevant subregions of a brain tumor that can be observed: the whole tumor, the tumor core, and the enhancing tumor. Our goal is to be able to classify each pixel of an MRI image into these tumor categories or background tissue. By leveraging multimodal MRI inputs and a deep neural network, this project aims to produce accurate tumor localization that could support clinical decision-making.

Our results show that the proposed U-Net model can effectively segment tumor regions from multi-modal MRI scans. The network achieves a validation Dice score of 0.8328, demonstrating strong overlap between predicted masks and expert-annotated ground truth labels. Training and validation losses decrease consistently throughout training, indicating stable optimization and good generalization. Qualitative visualizations further confirm that the model accurately captures the overall tumor structure and boundaries in MRI slices. These results demonstrate that deep convolutional encoder-decoder architectures such as U-Net are well-suited for automated medical image segmentation tasks and can provide reliable tumor localization for potential clinical applications.

2 Related Work

Deep learning has revolutionized the field of medical image segmentation. The foundational U-Net architecture [1] established the standard encoder-decoder paradigm with skip connections, which has since been adapted into frameworks like nnU-Net [2] that currently dominate the BraTS leaderboard. To address the severe class imbalance inherent in tumor segmentation, Milletari et al. introduced the Dice Loss function, which directly optimizes the intersection over union of predicted masks [3]. Furthermore, works by Havaei et al. have demonstrated the necessity of utilizing multimodal MRI inputs (T1, T2, T1CE, FLAIR) to capture distinct tumor subregions [4]. Our project builds upon these works by implementing a custom 4-level 2D U-Net from scratch, optimized with Dice Loss, to effectively perform multi-class localization across fused multimodal MRI inputs.

3 Methods

Our method employs a 2D U-Net architecture to segment brain tumor regions from axial MRI slices. U-Net is a fully convolutional network consisting of a contracting encoder path and an expanding decoder path connected by skip connections. The encoder progressively extracts hierarchical feature representations through convolution and downsampling operations, while the decoder upsamples these features to generate pixel-level predictions.

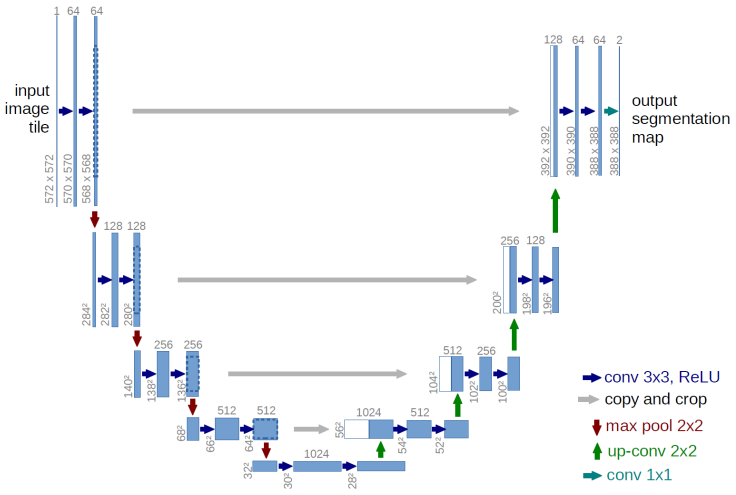

Figure 1: 2D U-Net Architecture Diagram [1]

We selected the U-Net due to its strong performance in medical image segmentation tasks, particularly when training data is limited. Its symmetric encoder-decoder architecture and skip connections enable the network to capture both global contextual information and fine-grained spatial details, which

are essential for accurately characterizing tumors that have irregular shapes and diffuse boundaries. Additionally, U-Net has a relatively small number of parameters compared to other classification networks, making it computationally efficient and practical for this project.

To optimize our network against severe class imbalance between tumor and background pixels, we utilized the Dice Loss function, formulated as

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_{i=1}^{N} p_i g_i + \epsilon}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} g_i + \epsilon}$$

where p_i represents the predicted probability of the tumor class for pixel i, g_i is the ground truth, and ϵ is a smoothing constant.

The primary strength of the U-Net architecture lies in its ability to perform precise localization while maintaining high accuracy. By combining multiscale contextual features with high-resolution spatial information, the network can effectively segment small or heterogeneous tumor regions. Furthermore, the design of U-Net allows it to be easily incorporated with pretrained encoder backbones or multimodal MRI inputs, making it a flexible framework that can be improved upon.

3.1 Training Algorithm

The training process is designed to optimize the network parameters by minimizing the Dice Loss over multiple iterations. We utilize a supervised learning approach where the 2D U-Net processes axial slices from the multimodal MRI volumes.

The procedure follows these steps:

- **Data Augmentation and Loading:** Input slices (T1, T1CE, T2, FLAIR) are normalized and fed into the model in mini-batches to ensure stable gradient descent.
- **Forward Pass:** The multimodal input $X \in \mathbb{R}^{H \times W \times 4}$ is passed through the encoder-decoder structure. The final layer applies a Sigmoid activation to generate a probability map $P \in [0, 1]$ for each of the target classes.
- **Loss Computation:** We calculate the Dice Loss ($\mathcal{L}_{Dice}$) between the predicted probability maps and the ground truth one-hot encoded masks. This mitigates the severe class imbalance where background pixels vastly outnumber tumor pixels.
- **Backpropagation:** Gradients ($\nabla_\theta \mathcal{L}_{Dice}$) are computed via backpropagation. We use the AdamW Optimizer to update the network weights (θ), adaptively tuning the learning rate for each parameter.
- **Model Selection:** At the end of each epoch, the model is evaluated on a disjoint validation set. We implement a best-model checkpoint strategy, where the model's state dictionary is saved only if the validation Dice Score improves, ensuring the final model generalizes well and avoids overfitting.

3.2 Training and Evaluation Algorithm

Once training is complete, the best-performing weights are loaded to evaluate the model on unseen data.

The testing pipeline is structured as follows:

- **Inference Mode:** The model is set to `eval()` mode, disabling dropout layers and freezing Batch Normalization statistics. The forward pass is wrapped in `torch.no_grad()` to reduce memory consumption and prevent accidental weight updates.
- **Pixel-wise Classification:** For each test slice, the model outputs a multi-channel probability map. We apply a thresholding operation (e.g., $p > 0.5$) across the channel dimension to convert probabilities into distinct, binary segmentation masks for each tumor subregion.
- **Evaluation Metrics:**
  - Quantitative: We calculate the Dice Coefficient for the whole tumor, tumor core, and enhancing tumor regions to provide a standardized measure of overlap.
  - Qualitative: We perform visual inspections by overlaying the predicted masks onto the FLAIR modality scans to verify if the model effectively captures the diffuse boundaries.

4 Experiments

The dataset used in this project is the publicly available BraTS (Brain Tumor Segmentation) dataset, which consists of multimodal magnetic resonance imaging (MRI) scans of patients with gliomas, including both adult and pediatric cases. Each patient scan includes four MRI modalities: T1, T1-weighted contrast-enhanced (T1CE), T2, and FLAIR. In addition to the imaging data, the dataset provides expert-annotated ground truth segmentation masks that describe tumor subregions such as the enhancing tumor, tumor core, and surrounding edema. All images in the BraTS dataset are provided in NIfTI format (.nii.gz) and are preprocessed to ensure consistency across subjects, as they are co-registered to the same anatomical template, interpolated to the same resolution (1 mm^3), and skull-stripped. The volumetric MRI data are processed into 2D axial slices, which are stored in HDF5 format to reduce memory usage. Each slice has metadata that includes the volume identifier, slice index, and corresponding target label.

The primary goal of the experiments we performed is to evaluate the effectiveness of a U-Net-based architecture for automated brain tumor segmentation. First, a baseline 2D U-Net model was trained to segment tumor regions from MRI slices, and performance was evaluated using the Dice coefficient. This experiment establishes a reference performance level and validates the overall segmentation pipeline. Our next experiment explored the impact of architectural design choices on segmentation performance. We performed experiments using different combinations of MRI modalities to analyze how multimodal inputs influence segmentation accuracy. Altogether, these experiments aim to demonstrate the clinical relevance of deep learning-based segmentation and to identify key factors to improve tumor localization accuracy.

4.1 Quantitative Results

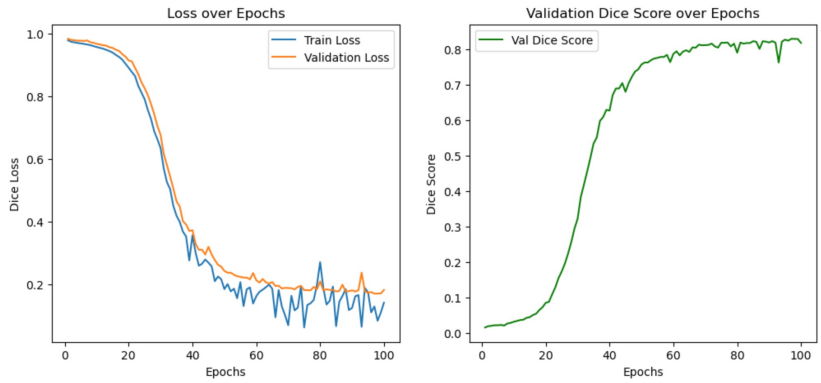

Figure 2: Loss and Dice Score Graphs over 100 Epochs

As shown in Figure 2, the model converges smoothly, reaching a peak validation Dice score of 0.8328 around epoch 80. The absence of a spike in validation loss indicates that the model did not severely overfit to the training data.

4.2 Brain Tumor Segmentation Results

Metric	Value
Training Loss	0.0769
Validation Loss	0.1672
Validation Dice Score	0.8328

Table 1: Final training loss, validation loss, and validation Dice score

Our final 4-level U-Net model successfully segmented tumor regions with an overall validation Dice Score of 0.8328, demonstrating the effectiveness of early multimodal fusion and Dice Loss optimization.

4.3 Qualitative Results

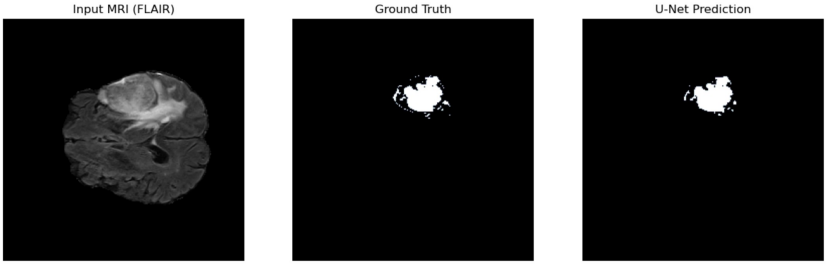

Figure 3: FLAIR, Ground Truth, and U-Net Prediction display for a brain containing a tumor

Figure 4: FLAIR, Ground Truth, and U-Net Prediction display for a healthy brain

Figure 3 illustrates a qualitative comparison between the U-Net prediction and the ground truth mask. The U-Net prediction successfully captures the overall shape and location of the tumor core, closely mirroring the ground truth mask. Minor artifacts in the background demonstrate the difficulty of segmenting diffuse boundaries, but the core localization is highly accurate.

Figure 4 presents an example of a healthy brain MRI slice with no tumor present. The model correctly predicts an empty segmentation, indicating that no abnormal tumor regions were detected. This result demonstrates that the U-Net does not simply predict tumor regions everywhere, but is capable of distinguishing between healthy tissue and pathological regions. The absence of false positives in this example suggests that the model has learned meaningful spatial and intensity features of tumors from the BraTS dataset, allowing it to generalize to normal brain structures without incorrectly labeling them as tumor tissue.

The model's convergence was monitored via Loss and Dice Score curves. As seen in our results, the training and validation losses decreased monotonically, suggesting a stable learning rate and effective optimization via the Dice Loss function. The gap between training and validation metrics remained narrow, indicating that the 4-level U-Net architecture provides sufficient regularization for the BraTS dataset.

4.4 Ablation Study

We claimed that early fusion of 4 MRI modalities (T1, T1CE, T2, FLAIR) was necessary. To prove the effectiveness of multimodal MRI fusion, we conducted an ablation study training the U-Net solely on FLAIR scans , where we set in_channels=1. The single-modality model achieved a Dice score of 0.7755, which is lower than our 4-channel model result of 0.8328, but is still a high Dice score for the whole tumor region. When analyzing the U-Net prediction for this study (Figure 4), we can see that it lacks features in the internal subregions. which are the tumor core and enhancing tumor parts. This study confirms that while single-modality FLAIR is sufficient for rough localization (Whole Tumor), early fusion of all four modalities is still necessary for a more nuanced multi-class subregion segmentation required for surgical planning.

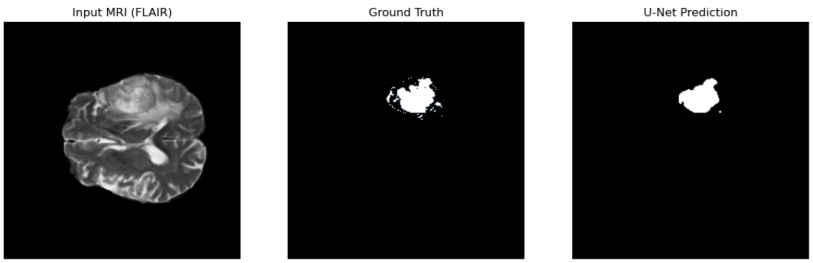

Figure 5: FLAIR, Ground Truth, and U-Net Prediction iisplay for modality ablation

5 Supplementary Material

The corresponding codes are made publicly available in GitHub [6]. A supplementary video presentation is available as well [7].

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